

# LAB EXPERIMENTS

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36. Write a Python program to draw a scatter plot for three different groups comparing weights and heights.

**Code :**

```
import matplotlib.pyplot as plt

# Data for three different groups
group1_weight = [55, 56, 59, 60, 62, 63, 64, 66]
group1_height = [115, 125, 124, 135, 150, 165, 124, 135]

group2_weight = [61, 62, 63, 64, 67, 68, 69, 70]
group2_height = [158, 195, 180, 153, 170, 165, 192, 235]

group3_weight = [71, 71.5, 72, 72, 70.5, 68.5]
group3_height = [180, 175, 135, 177, 160, 135]

# Scatter plot for three groups
plt.scatter(group1_weight, group1_height, label="Group 1", marker="o")
plt.scatter(group2_weight, group2_height, label="Group 2", marker="s")
plt.scatter(group3_weight, group3_height, label="Group 3", marker="^")
```

```
# Labels and title
```

```
plt.xlabel("Weight")
```

```
plt.ylabel("Height")
```

```
plt.title("Group wise Weight vs Height scatter plot")
```

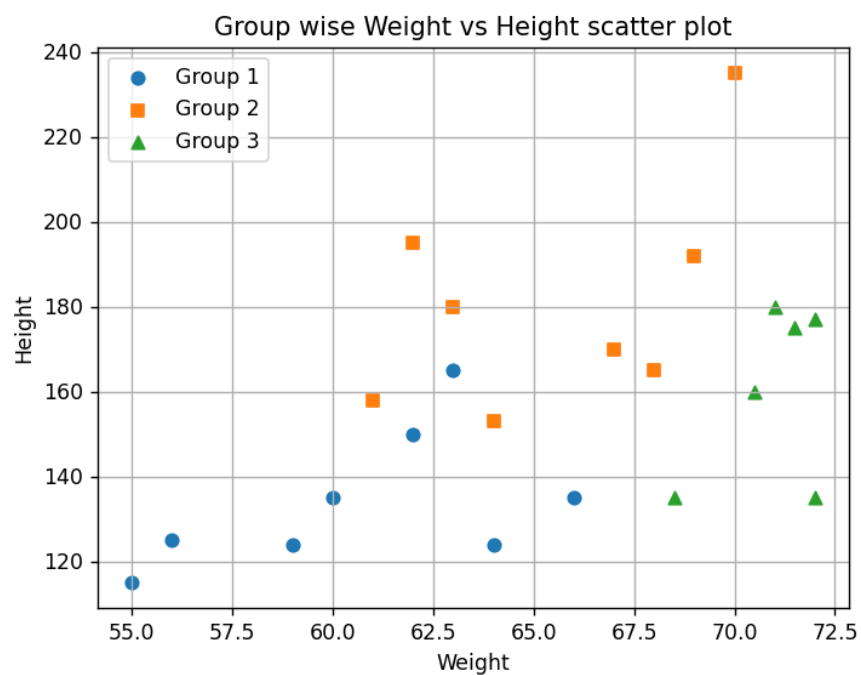
```
# Display legend and grid
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

**OUTPUT :**



37. Write a Pandas program to create a dataframe from a dictionary and display it.

Sample data: {'X': [78, 85, 96, 80, 86],  
'Y': [84, 94, 89, 83, 86], 'Z': [86, 97, 96, 72, 83]}

### Code :

```
import pandas as pd
# Creating a dictionary
data = {
    'X': [78, 85, 96, 80, 86],
    'Y': [84, 94, 89, 83, 86],
    'Z': [86, 97, 96, 72, 83]}
# Creating a DataFrame
df = pd.DataFrame(data)
# Displaying the DataFrame
print(df)
```

### OUTPUT :

```
PS E:\Query process\lab> e;; cd 'e:\Query process\lab'; & 'C:\Python312\python.exe' 'c:\Users\Vedvyass M\.vscode\extensions\ms-python\launcher' '49716' '--' 'E:\Query process\lab\37'
   X  Y  Z
0  78  84  86
1  85  94  97
2  96  89  96
3  80  83  72
4  86  86  83
```

38. Write a Pandas program to create and display a DataFrame from a specified dictionary data which has the index labels.

Sample Python dictionary data and list labels:

```
exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
            'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

**Code :**

```
import pandas as pd
import numpy as np

# Dictionary data
exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
            'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no',
                'yes', 'yes', 'no', 'no', 'yes']}
```

```
# Index labels
```

```
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

```
# Create DataFrame with index labels
```

```
df = pd.DataFrame(exam_data, index=labels)
```

```
# Display DataFrame
```

```
print(df)
```

**OUTPUT :**

```
PS E:\Query process\lab> e;; cd 'e:\Query process\lab'; & 'c:\Program Files\Python312\python.exe' 'c:\Users\Vedvyass M\.vscode\extensions\ms-python.python\python\launcher' '59718' '--' 'E:\Query process\lab\38'
```

|   |           |      |   |     |
|---|-----------|------|---|-----|
| a | Anastasia | 12.5 | 1 | yes |
| b | Dima      | 9.0  | 3 | no  |
| c | Katherine | 16.5 | 2 | yes |
| d | James     | NaN  | 3 | no  |
| e | Emily     | 9.0  | 2 | no  |
| f | Michael   | 20.0 | 3 | yes |
| g | Matthew   | 14.5 | 1 | yes |
| h | Laura     | NaN  | 1 | no  |
| i | Kevin     | 8.0  | 2 | no  |
| j | Jonas     | 19.0 | 1 | yes |